

ELECTRICAL SAFETY AND ENERGY AUDIT REPORT
FOR
INDIAN BANK – PALARIVATTOM BRANCH
(2025-2026)



Prepared by
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	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
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	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

TABLE OF CONTENTS

	PAGES
ABBREVIATIONS	4
1 EXECUTIVE SUMMARY.....	5
2 BASIC TERMINOLOGY	6
3 TYPES OF ELECTRICAL INJURIES.....	8
4 INTRODUCTION	9
5 SCOPE FOR ELECTRICAL AUDIT AND ENERGY AUDIT	10
6 METHODOLOGY	11
7 EARTHING RESISTANCE:	12
8 LOAD MEASUREMENT STUDY	12
9 THERMOGRAPHY	14
10 TOTAL LIGHTING CONNECTED LOAD	18
11 CONCLUSION	20
APPENDIX 1 – ELECTRICAL SAFETY AND ENERGY AUDIT REPORT - CHECKLIST	22

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

ABBREVIATIONS

ACB	AIR CIRCUIT BREAKER
CEA	CENTRAL ELECTRICITY AUTHORITY
CT	CURRENT TRANSFORMER
DG	DIESEL GENERATOR
ELCB	ELECTRIC LEAKAGE CIRCUIT BREAKER
ESA	ELECTRICAL SAFETY AUDIT
FE	FIRE EXIT
HAC	HAZARDOUS AREA CLASSIFICATION
IE	INSPECTOR ELECTRICAL
IEC	INTERNATIONAL ELECTROTECHNICAL COMMISSION
IR	INSULATION RESISTANCE
LT	LOW TENSION PANELS
MCB	MINIATURE CIRCUIT BREAKERS
MCC	MOTOR CONTROL CENTER
NPC	NATIONAL PRODUCTIVITY COUNCIL
PA	PUBLIC ANNUNCIATION
PT	POTENTIAL TRANSFORMER
UPS	UNINTERRUPTED POWER SUPPLY
UPVC	UNPLASTICIZED POLYVINYL CHLORIDE
XLPE	CROSS LINKED POLYETHYLENE CABLE

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT - PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

1 EXECUTIVE SUMMARY

This Electrical Safety Audit Report presents a comprehensive evaluation of the electrical systems and safety measures at Indian Bank, Ernakulam zone –Palarivattom branch. The audit was conducted on 19/05/2026 in accordance with NEC-2023 guidelines, CEA Regulations 2023, Indian Bank Standards and relevant Indian Standards.

The audit team has documented all observations related to Electrical safety, which are now submitted for review to the management of M/s. Indian Bank- Ernakulam Zone, these findings are intended to guide the organization in strengthening its Electrical Safety Framework of the Branches and Offsite ATM's.

Audit Dates: 19/05/2026

Location: Palarivattom branch

Audit Conducted By: M/s. Aura Veritas Engineering and Consultancy Pvt. Ltd.

1.1 Key Findings:

The overall electrical infrastructure of the branch was found to be Operational with several areas requiring immediate attention and corrective actions to ensure compliance with electrical safety standards and to prevent potential hazards.

1.2 Overall Assessment:

The electrical systems require **Minor to Medium level improvements** to meet **NEC-2023** Indian Bank Standards compliance standards and to ensure safety of personnel, equipment, and continuity of banking operations.

This report details all observations with photographic evidence, technical measurements, and prioritized recommendations for corrective actions.

The audit encompassed a comprehensive evaluation of electrical safety practices, including:

- Electrical Maintenance Team One to One Interview during Audit days.
- Examination of Electrical Equipment at premises.
- Verification of Electrical Parameters such as Load Measurements, Earthing Measurements, Lux levels measurements, Maximum Current ratings verification, Thermography measurements at each panel and Distribution boards.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT - PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

- Direct Testing of Fire and Smoke Sensors, Fire Alarm panels in the facility.
- Validation of Electrical Equipment for critical and risky areas.
- Assessment of Maintenance Records related to Electrical safety.
- Inspection of all relevant Electrical Engineering and Maintenance Documents.
- Evaluation of Compliance with Previous Audit Reports
- Benchmarking against NEC-2023 Electrical Safety Guidelines and Branch Checklist for the Electrical Safety.

2 BASIC TERMINOLOGY

1. Key Electrical Engineering Terms Everyone Should Know

Understanding these terms is crucial for anyone working with electrical systems, whether you're conducting audits, managing installations, or ensuring safety compliance.

2. Basic Electrical Quantities

- **Voltage (V):** The electrical potential difference between two points. Measured in volts.
- **Current (I):** The flow of electric charge. Measured in amperes (A).
- **Resistance (R):** Opposition to current flow. Measured in ohms (Ω).
- **Power (P):** The rate at which electrical energy is consumed or generated. Measured in watts (W).
- **Energy (E):** Total power consumed over time. Measured in kilowatt-hours (kWh).

3. Components & Devices

- **Circuit Breaker:** A safety device that interrupts current flow in case of overload or fault.
- **Transformer:** A device that changes voltage levels between circuits.
- **Capacitor:** Stores electrical energy temporarily and releases it when needed.
- **Inductor:** Stores energy in a magnetic field when current flows through it.
- **Relay:** Electrically operated switch used for control and automation.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT - PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

4. System Concepts

- **Single-phase vs. Three-phase:** Types of AC power systems; three-phase is more efficient for industrial use.
- **Earthing (Grounding):** Connecting electrical systems to the earth to prevent shock hazards.
- **Load:** The amount of electrical power consumed by devices or systems.
- **Short Circuit:** An abnormal connection causing excessive current flow, often leading to damage or fire.
- **Overload:** When a circuit carries more current than it's designed for,

5. Measurement & Safety

- **Insulation Resistance:** Measures the effectiveness of insulation in preventing current leakage.
- **Thermography:** Infrared imaging used to detect hot spots in electrical systems.
- **Harmonics:** Distortions in electrical waveforms that can affect equipment performance.
- **RCD (Residual Current Device):** Detects leakage currents and disconnects power to prevent electrocution.
- **Lux Level:** Measurement of light intensity, relevant in lighting audits.

6. Codes & Standards

- **IS Codes:** Indian Standards for fire safety and protection.
- **NEC-2024** Guidelines
- **BS EN:** British/European standards for fire equipment and systems.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

3 TYPES OF ELECTRICAL INJURIES

Types of Injuries from Electrical Incidents



Electric Shock

A reflex response possibly involving trauma which occurs when electrical current passes over or through a worker's body. It usually involves burns and abnormal heart rhythm and unconsciousness.



Electrocution

Electrocution occurs when electrical current passes over or through a worker's body resulting in a fatality.



Falls

Electric shock may cause muscles to contract causing a worker to lose his or her balance and fall. An explosion from an electrical incident can also cause a fall.



Burns

Electrical burns are the most common shock-related, nonfatal injury. They occur when a worker contacts energized electrical wiring or equipment. Although electrical burns can occur anywhere on the body, they most often occur on the hands.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

4 INTRODUCTION

Indian Bank is a trusted banking leader with strong national and regional presence, offering comprehensive financial services and robust ATM coverage for convenient banking. Branches and ATMs in Ernakulam serve local customers with wide-ranging solutions and digital support.

4.1 AIMS AND OBJECTIVES OF ELECTRICAL SAFETY & ENERGY AUDIT

- Verify Electrical Safety of the Indian Bank branches developed/implemented in accordance with system safety program plan requirement as per NEC-2023 Guidelines and NBC-2016 Guidelines.
- Electrical Audit includes testing and verification of input from Mains, Input and Output of UPS, Condition of UPS, AC wiring and functioning of AC units, all wires, fuse carriers, MCBs Earth lighting and other circuits, Generator etc. Tests as per the format area are to be completed for each branch.
- Assess effectiveness of the agency's system Electrical safety systems.
- Identify potential Safety hazards in the operational system and weaknesses in the system safety programs.
- Verify prior corrective actions are being tracked for closure.
- Recommend improvements to the system safety program.
- Provide management with assessment of status and adequacy of system safety program.
- Assure continuing evaluation of safety related issues, awareness and reporting.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

5 SCOPE FOR ELECTRICAL AUDIT AND ENERGY AUDIT

Electrical Safety Audit Scope in Compliance with NEC-2023 & IS Standards

1. Standards & Compliance

- Conduct audit in accordance with **NEC-2023, NBC-2016** relevant **Indian Standards**
- Ensure alignment with statutory and regulatory electrical safety norms and BEE Norms.

2. Preventive Maintenance Review

- Evaluate the Electrical Preventive Maintenance (EPM) system
- Check for schedule adherence, documentation, and effectiveness.

3. Incident & Accident Review

- Analyse past Electrical accidents and incidents within the facility.
- Identify root causes and assess corrective/preventive measures taken.

4. Procedure & Documentation Review

- Review **Electrical Safety System procedures and Energy Management Procedures** for adequacy and implementation.
- Examine key documents and Conduct Electrical Safety Tests:
 - Register of Electrical Installations
 - Test Conducted:
 - Insulation Resistance Test
 - Earth Resistance Test
 - LT Panel Equipment Tests
 - Data Sheets of critical electrical installations

5. Thermography Analysis

- Review and Conduct **Thermography test results** of all electrical installations.
- Identify hotspots, loose connections, and thermal anomalies.

6. Emergency Power Systems

- Assess **supply reliability** and readiness of emergency systems:

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

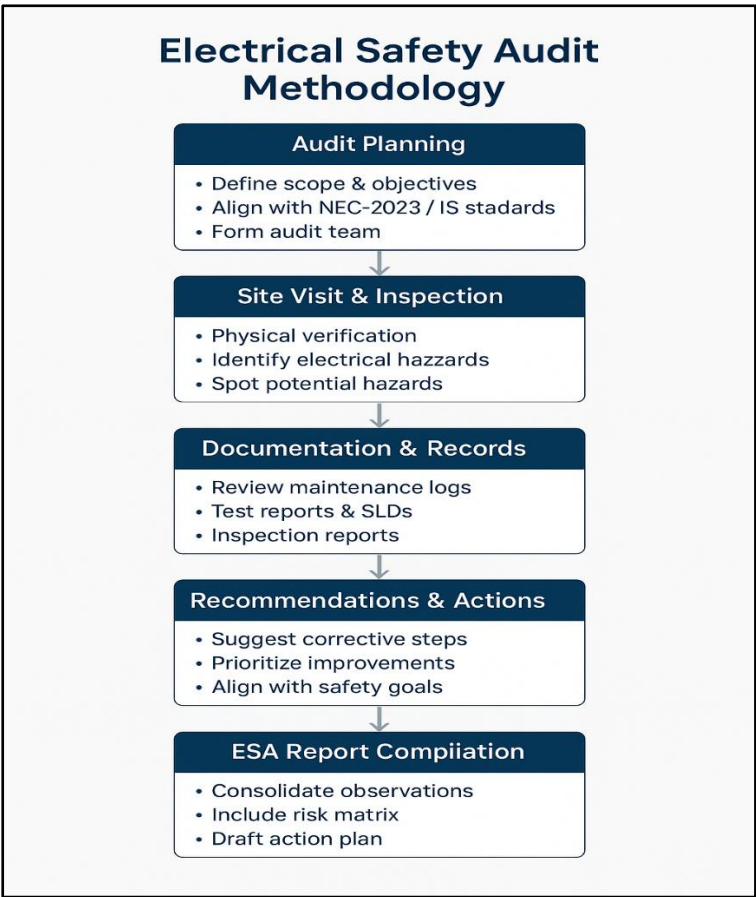
- Uninterruptible Power Supply (UPS)
- Battery Backup Systems

7. Report & Recommendations

- Compile findings into a **comprehensive audit report**.
- Provide **actionable recommendations** for safety and operational improvements.

6 METHODOLOGY

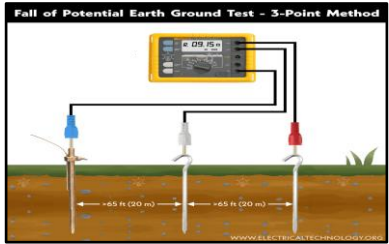
The Electrical Safety and Energy Audit include the review of documentation with regard to the scope covered in audit, an on-site visit, including Electrical Safety Measures, Energy Saving measures, and data collection, their review and analysis. Following methodology in detail:



	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

7 EARTHING RESISTANCE:

Earthing resistance is the **resistance between the grounding electrode and the surrounding soil**. It determines how effectively fault currents, lightning surges, or leakage currents are dissipated into the earth.



Important for all Applications:

Safety: Prevents electric shock hazards by directing fault currents safely into the ground.

Equipment Protection: Reduces the risk of damage to electrical systems during faults or lightning strikes.

Voltage Stabilization: Helps maintain consistent voltage levels across the system.

Ideal Resistance Values:

Application	Recommended Resistance
Power stations & substations	$\leq 1 \text{ ohm}$
Commercial buildings	$\leq 5 \text{ ohms}$
Residential systems	$\leq 10 \text{ ohms}$
NEC standard (if not achievable)	$\leq 25 \text{ ohms}$

8 LOAD MEASUREMENT STUDY

Load Flow Analysis is a computational method used to determine the steady-state operating conditions of an electrical power system. It evaluates how electrical power—both real (P) and reactive (Q)—flows through the network under specific load conditions.

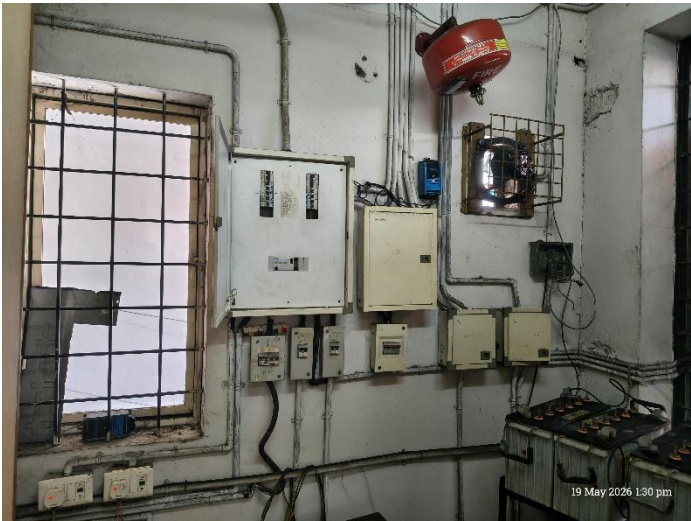
Important for all Applications:

- Required by NEC 220.87 for determining existing loads before adding new equipment
- Prevents overloads, phase imbalance, and fire hazards
- Ensures panel boards and feeders operate within safe limits
- Identifies peak demand periods and power wastage
- Supports Power factor correction and efficiency improvements

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

- Detects the Irregular load distribution, harmonics and voltage drop
- Avoid Cost retrofits

8.1 SITE AUDIT PICTURES



	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

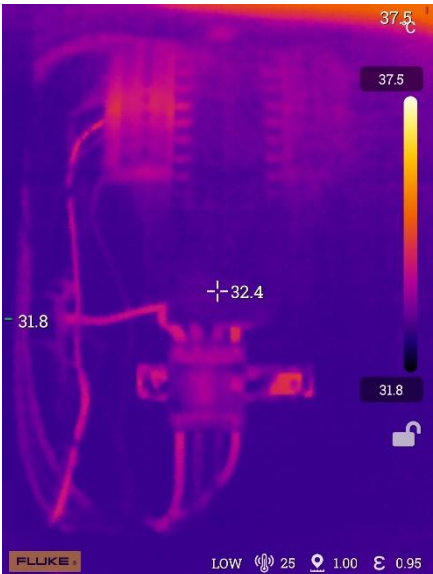
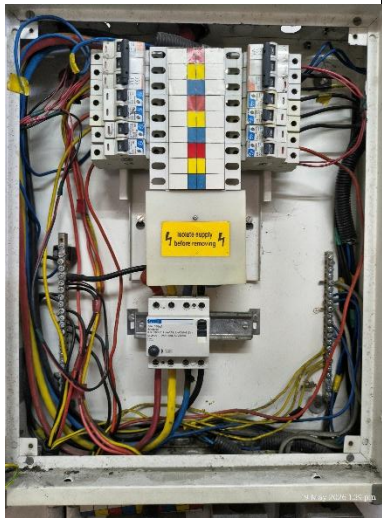
9 THERMOGRAPHY

Thermal imaging is a sophisticated and non-invasive technique that utilizes infrared technology to detect heat emissions from various objects. It converts infrared energy into a visible light display, helping detect abnormal heat patterns caused by overloads or impending equipment failures.

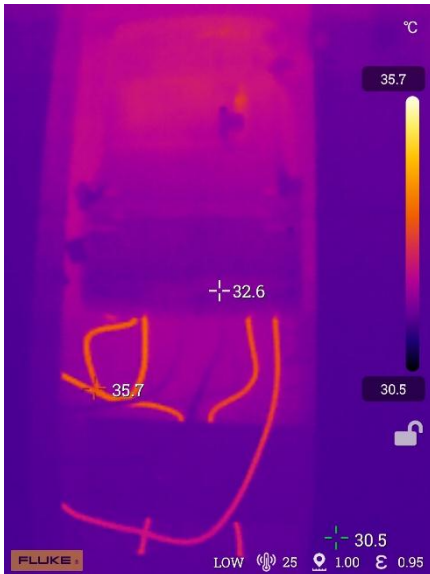
Important for all Applications:

- Preventing arc flash, insulation failure, and fire hazard
- Detecting hot spots in panels, breakers, and busbars
- Identifying loose connections and phase imbalance
- Monitoring transformer heating and motor bearings

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

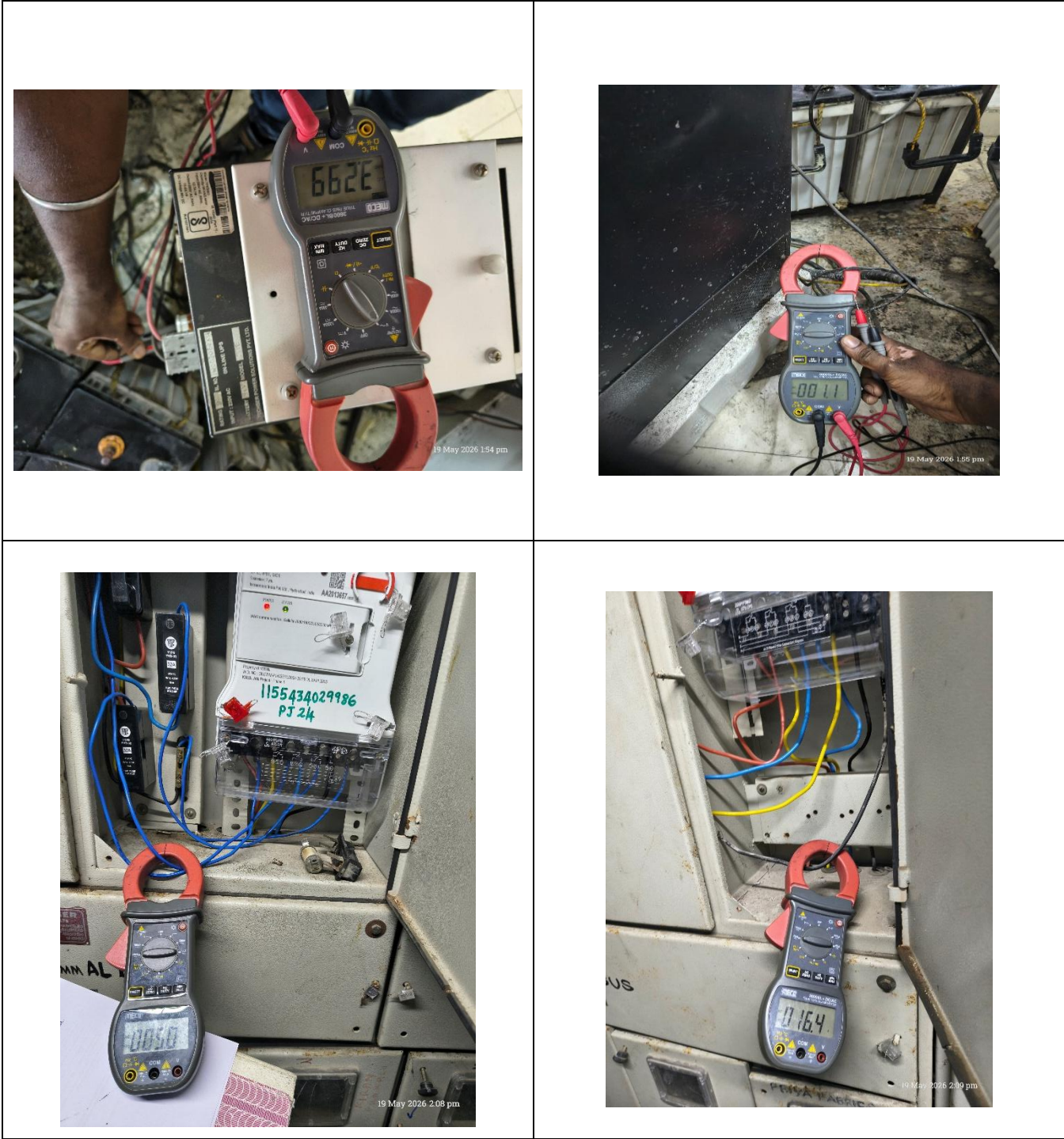
Thermography Hotspots during Site Audit			
Thermal pic	Actual pic	Location	Observation
		Fuse Hot spot = 39.5 °C	Normal
		Incoming Hot spot= 32.4°C	Normal

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

		Distribution boxes Hot spot = 33.2 °C	Normal
		EB meter Hot spot = 35.7 °C	Normal

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

9.1 SITE AUDIT PICTURES



	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

10 TOTAL LIGHTING CONNECTED LOAD

Lighting connected loads refer to the total electrical load attributed to lighting fixtures and systems within a facility.

Checklist during our Inspection:

- Verify fixture type and wattage
- Cross-check with lighting layout drawings
- Measure actual load using clamp meter
- Compare with design load and NEC/IS standards
- Identify retrofit opportunities for energy saving

Lighting Load for M/s. Indian Bank – Ernakulam zone - Palarivattom branch:

Observation:

During the Site Audit and Recording the Actual Loads at all Rooms, we are Satisfactory with the Results obtained from the Clamp meter and Engineering Predictions. We also found that majority of the Lighting Loads found **Acceptable**.

Opportunity for Improvement:

Based on the Site Audit findings and anomalies found during the normal operation please follow the below basic steps to increase the Efficiency of the Equipment and Proper Load distribution across all Common rooms.

- Old lights are to be replaced with Energy efficient Lights to minimize the Load during Peak hours.

Table 1: Phase load

R (kW)	Y (kW)	B (kW)
4.004	0.0638	0.363

Whether load is balanced or unbalanced? : YES

If load is unbalanced, any rectification to be done? : Load unbalance observed

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

Table 2: Number of electrical items at premises

Rooms	Tube light	Fan (wall)	Light 1	Light 2	PC	Printer	Cash Counting machine	AC	Exhaust fan	UPS
Banking hall	5	3	5		5	4	2	1		
Customer's hall		2	3					2		
Strong room	5	2								
Manager's Cabin			1	2	1					
Server room				2				1		
Staff room	1	1		2						
Canteen										
Stationery room										
Toilets	3								2	
UPS room										2
ATM			1	1				2		1
Total	14	8	10	7	6	4	2	6	2	3

Table 3: Total electrical load (in kW) at premises

Rooms	Tube light	Fan	Light 1	Light 2	PC	Printer	Cash Counting machine	AC	Exhaust fan	UPS
Power per item (W)	24	70	35	18	250	250	75	-	46	3000
Banking hall	120	210	175	0	1250	1000	150	4000	0	0
Customer's hall	0	140	105	0	0	0	0	2722	0	0
Strong room	120	140	0	0	0	0	0	0	0	0
Manager's Cabin	0	0	35	36	250	0	0	0	0	0
Server room	0	0	0	36	0	0	0	1361	0	0
Staff room	24	70	0	36	0	0	0	0	0	0
Canteen	0	0	0	0	0	0	0	0	0	0
Stationery room	0	0	0	0	0	0	0	0	0	0
Toilets	72	0	0	0	0	0	0	0	92	0
UPS room	0	0	0	0	0	0	0	0	0	6000
ATM	0	0	35	18	0	0	0	2722	0	1000
Total item wise	336	560	350	126	1500	1000	150	10805	92	7000
Total power consumption in the premises in kW				21.919						

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

11 CONCLUSION

Based on the Site findings, technical document verification, Validation of test reports, Our Site Testing (Earth Resistance, Thermography, Load Imbalance Study assessment, Lightning Load Assessment, Lux Study Assessment) we found the below mentioned recommendations are to be taken care in order to maintain the NEC-2023 Guidelines, BEE Standards and Electrical Equipment & Personnel Safety. Final conclusions/recommendations are listed in Appendix 1.

S No.	Risk	Audit Observations/Shortcomings	Detailed Scope of Work for Rectifying the Observations/Shortcomings
1.	Medium	Un balanced load	Common Neutral Line Cable Temperature is High is due to Harmonic currents raised due to Variable loads such as Computers, Air Conditioners etc, In order to avoid this scenario we recommend to reduce the R-Phase load and Re-distribute to other Phases, Especially Group all Non-Linear Loads and supply them via separate Busbars, In case the Problem persists after lining all Non-Linear loads, change the Neutral Cable Size to the Next Cross sectional size to handle the Additional Non-Linear Current and Heat dissipation.
2.	Medium	Earth Pit maintenance	It is recommended to conduct a thorough inspection of earth electrodes and connections to check for rust, corrosion, or physical damage. Engage a certified third-party consultant to measure the earth resistance using a calibrated earth tester, ensuring that all readings comply with standard safety requirements.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

S No.	Risk	Audit Observations/Shortcomings	Detailed Scope of Work for Rectifying the Observations/Shortcomings
3.	Low	UPS output connection	UPS connection split up loads to be properly maintained as per emergency load distribution matrix
4.	Low	UPS Battery	UPS battery required scheduled maintenance.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

APPENDIX 1 – ELECTRICAL SAFETY AND ENERGY AUDIT REPORT - CHECKLIST

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

ELECTRICAL SAFETY AND ENERGY AUDIT REPORT -CHECKLIST

BRANCH DETAILS

S. No.	Description	Details
1	Zone Name	Ernakulam
2	Branch Name & Branch Code	Palarivattom branch,
3	Address	30/2335 B & B1, V E arcade, Opp. Vyapar Mandir, Palarivattom, Ernakulam District-682025
4	Sanctioned load	24 kW
5	Connected Load	23.36 kW
6	Monthly Electricity bill (Appx)	INR 20,000/- Approx
7	Total tonnage of Air conditioner load	10 Tons
8	Audit Date	19/05/2026

ELECTRICAL ENERGY AUDIT

S. No.	Description	Quantity	Recommendation
1	Type of Electrical Fittings installed and its	NIL	
2	No of Ceiling Fan installed and its power consumption.	7 and 490W	
3	No. of Wall Fan installed and its power consumption	2 and 140W	
4	No of Air-conditioners and its capacity and power consumption of each.	6 nos. 10 tons and 10.8 kW	
5	No of computer installed and its power consumption	06 and 1500 W	
6	UPS installed and its capacity	7 (3+3+1) KVA & 3 (1+1+1) nos	
7	Any other gadgets installed and its power consumption.	-	
8	Verification of electricity Bill for last 12 months and observation if any	Found fine	
9	Leakage if any in the wiring system	NIL	

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

INSPECTION DETAILS

S.No.	Description	Site Observation
1.	Condition of Main Distribution Board / Sub Distribution Board.	Good
2.	Cables & wires used in the installation to be checked for required insulation and current carrying capacity. Condition of wiring with specific remark Note: Bunching of wires to be avoided & Proper clamping of conduits with wires to be done in the wall.	Good
3.	Loose connection & hanging wires	No hanging wires
4.	Heavy duty contact points to be checked for the firm connection to avoid overheating and minimize power loss	Checked and appears well
5.	Condition of fuses/main switches Note: If branches are provided with old switches and boards it can be replaced with MCBs and distributions boards of suitable capacity	Up to the Standards
6.	Whether any single plug point used for various power applications by means of any extension board Note: for avoiding overloading of circuits	No extension board utilized
7.	Non provision of MCB / Circuit Breakers of adequate capacity. Note: Incoming and individual outgoing MCBs/switched to be rated with proper capacity	YES
8.	Condition of neutral wire / earthing Note: check for any loose connection	Good condition
9.	Details of connected load	25 kW
10.	Sanctioned Load (By State Electricity Board) (in KW) Present Load (in KW) If the available Load is more than the sanctioned load then the additional equipment which are connected are to be ensured that the wiring / switches / MCBs provided is capable of withstanding additional load. This is the root cause of fire accidents.	23.36 kW
11.	Power Consumption in excess of the sanctioned load.	No energy meter
12.	Check for availability of measuring instruments in the Control Panel viz. Voltmeter, Ammeter, Frequency Meter,	Only EB Meter No Energy Meter

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

S.No.	Description	Site Observation
	Power Factor Meter, Energy Meter for its calibration to measure electricity.	
13.	Earthing: (a) UPS Earthing – Value in ohms. (b) Raw power Earthing – Value in ohms. (c) No of earth pits Note: UPS and raw power should have separate earthing	(a) 0.32 volts (b) Above 50 ohms (2.35 volts) (c) not identified
14.	Total UPS Load: (a) Total no of UPS with capacity (b) UPS panel wiring (c) Whether alternate Nodes are connected to one UPS. (d) Is UPS under AMC with OEM or not? (e) Is UPS load balanced? Mention current in all the three phases Note: If UPS load is imbalanced, the loads to be balanced in all the three phases	(a) 7 (3+3+1) KVA – 3 (1+1+1) nos (b) Good (c) No (d) YES (e) Single phase UPS
15.	UPS Batteries: (a) No of Batteries (b) Periodicity of servicing (c) Proper ventilation / exhaust fan provided	(a) 19 no's (16+3) (b) OEM recommended (c) YES
16.	Air-conditioner: (a) No. of Air conditioners working round the clock with Tonnage. (b) Whether separate stabilizer provided for each AC unit. (c) Any timer fixed to the AC units for alternate functioning. (d) Check whether supply for ACs is drawn through power points. (e) Date of last servicing carried out. (f) Eligible life span of ACs. (g) Whether Outdoor units are earthed? (h) Whether AC's are covered in AMCs? If Yes, what is the frequency of Preventive Maintenance in a year? Note: Periodic cleaning of AC filters is mandatory.	(a) 5 (4 Tons = 1, 2 Tons = 1, 1 ton = 3) (b) YES (c) NO (d) YES (e) Bill not available. (f) – (g) Not Identified (h) YES, But bill not provided
17.	Any Generator provided.	No Separate DG for Bank.

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

SAFETY QUESTIONNAIRE

S.NO.	DESCRIPTION	DETAILS	REMARKS
1	Whether TPN Isolators / MCCBs / ELCBs / are provided to cater the load	☒ YES / ☐ NO	
2	Whether lights and emergency lights are provided in electrical room / operating areas for easy operation & maintenance works	☒ YES / ☐ NO	
3	Whether pump room, dg set room, ups room, ac plant room, electrical panel room are maintained in dry and in good condition and obsolete / hazardous / old items are not dumped there.	☐ YES / ☒ NO	
4	Whether water seepage is observed near any of the electrical panel, distribution board, electrical equipment, etc	☐ YES / ☒ NO	
5	whether earthing pits are provided and connected to equipment and the body of equipment	☒ YES / ☐ NO	
6	Whether earthing pits are maintained properly	☐ YES / ☒ NO	Earth pit to be maintained properly
7	Whether exhaust fans for ventilation of panel room / electrical room / UPS room / DG set room is provided and no paper, old material or any other scrap is kept near DB / panels / ups / batteries etc.	☒ YES / ☐ NO	
8	Whether penalty is being imposed in electricity bill on account of higher load / poor power factor / poor load factor etc (ascertained from the electricity bills of april/may/june/july) additional electrical load required (if any)	☐ YES / ☒ NO	
9	Whether load is distributed in all three phases to avoid any unbalancing of phase load and no loose electrical connections /	☐ YES / ☒ NO	

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

S.NO.	DESCRIPTION	DETAILS	REMARKS
	haphazrad wirings observed in the building / office premises.		
10	Whether isolating switches are provided for switching off of the non essential loads in premises during night and main switch to switch off the power supply to the building / floor in case of fire / emergency.	☒ YES / ☐ NO	Main LDB available
11	Whether proper preventive maintenance after opening of the electrical boards and distribution boards are carried out by the license holder electrician or skilled technicians of equipment manufacturer / service providers.	☒ YES / ☐ NO	
12	Whether appropriate timers used in the changeover of air conditioners for server room ACs and for signage boards to make auto on / off (for schedule timings). the thermostat of the ACs at server room should be set to say 26°C, so that it will run only when the temperature is too high (to minimize chances of fire due to idle running of the ACs during night)	☐ YES / ☒ NO	One AC not working condition
13	Whether preventive maintenance of the electric installation and equipment is carried out by skilled license holder electricians / skilled technicians	☒ YES / ☐ NO	
14	General condition of electrical control panels, main switch, electric meter board, change over switch, AC, water coolers, water filters, wiring cables etc are in good conditions and DBs, panels, switch boards are properly covered	☒ YES / ☐ NO	
15	Whether electrical safety equipments of pantry etc are properly connected to industrial socket box with MCBs or latest type switches are provided to switch on / off the ACs to protect them from overload.	☒ YES / ☐ NO	

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

S.NO.	DESCRIPTION	DETAILS	REMARKS
16	Whether the contact number of persons, electricians, power distribution company, generator service provider, ups vendors, ACs etc are available with security guards and other staff are displayed in electric / UPS / server room.	☐ YES / ☒ NO	
17	Whether power factor correction panel of appropriate rating is installed.	☒ YES / ☐ NO	
18	Whether all the connecting point at various DBs and panels are properly insulated and proper indicative marking are done at the respective panel / DB	☐ YES / ☒ NO	Insulation and maintenance not provided and performed in end terminal panel
19	Whether the internal insulation of the wire have been accounted for checking the durability of electrical connection	☐ YES / ☒ NO	
20	All old disposable records, broken furniture etc accumulated in the premises have been cleared.	☐ YES / ☒ NO	Kept in store room
21	Combustible leaf litter / waste papers etc in and around the branch are removed / cleaned periodically	☒ YES / ☐ NO	
22	No stationary / records / old obsolete items are stored / kept in the server / UPS / electric room	☒ YES / ☐ NO	
23	Storage racks in stationary / record room kept at a safe distance of at least 2m from electrical points / switch / junction boxes	☒ YES / ☐ NO	
24	Whether in the pantry / canteen, LPG is used	☐ YES / ☒ NO	
25	Are the fire extinguishers available in the following work area and clearly marked and accessible: 1.Server / ups room: CO2 type x 2	☒ YES / ☐ NO	

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH				
	Document No		AVPL/IBE/25-26/ESEA-05		
	Date of Report Issue	30/05/2026	Rev. No.	0	

S.NO.	DESCRIPTION	DETAILS	REMARKS
	2. Individual department of the premises: CO2 type x 1 3. Stationary room: CO2 type x 1 4. Electric room / ac plant room: CO2 type x 2 5. DG set/generator: 6kg ABC capacity x 2		
26	Whether server room have dual ac units with timer circuit device on independent	☐ YES / ☒ NO	
27	Whether exhaust fans are installed in UPS room	☒ YES / ☐ NO	
28	Whether proper insulation at the batteries have been provided	☒ YES / ☐ NO	Batteries need to be maintained

UPS DETAILS

Parameters Checked	Readings Recorded at UPS 1	Readings Recorded at UPS 2	Readings Recorded at UPS 3 (ATM)	Normal Range	Remarks
Voltage at Input of UPS (V) (Vac)	251	244	230	230 +/- 10%	
Voltage at Output of UPS (V) (Vac)	230	230	230	230 +/- 10%	
Load on UPS (KVA) Numeric	3	3	1	-	
Frequency of Supply (Hz) at O/P of UPS	50	50	50	50 +/- 5%	
Battery Voltage (V)	108	108	40.5	-	
Battery Rating Type	12V / 100AH		12V / 100AH		
Total Battery	16 nos		3 nos		

	ELECTRICAL SAFETY AND ENERGY AUDIT REPORT – PALARIVATTOM BRANCH			
	Document No		AVPL/IBE/25-26/ESEA-05	
	Date of Report Issue	30/05/2026	Rev. No.	0

AUDIT OBSERVATIONS/SHORTCOMINGS

S No.	Risk	Audit Observations/Shortcomings	Detailed Scope of Work for Rectifying the Observations/Shortcomings
5.	Medium	Un balanced load	Common Neutral Line Cable Temperature is High is due to Harmonic currents raised due to Variable loads such as Computers, Air Conditioners etc, In order to avoid this scenario we recommend to reduce the R-Phase load and Re-distribute to other Phases, Especially Group all Non-Linear Loads and supply them via separate Busbars, In case the Problem persists after lining all Non-Linear loads, change the Neutral Cable Size to the Next Cross sectional size to handle the Additional Non-Linear Current and Heat dissipation.
6.	Medium	Earth Pit maintenance	It is recommended to conduct a thorough inspection of earth electrodes and connections to check for rust, corrosion, or physical damage. Engage a certified third-party consultant to measure the earth resistance using a calibrated earth tester, ensuring that all readings comply with standard safety requirements.
7.	Low	UPS output connection	UPS connection split up loads to be properly maintained as per emergency load distribution matrix
8.	Low	UPS Battery	UPS battery required scheduled maintenance.